SOLVED PRACTICE WORKBOOK

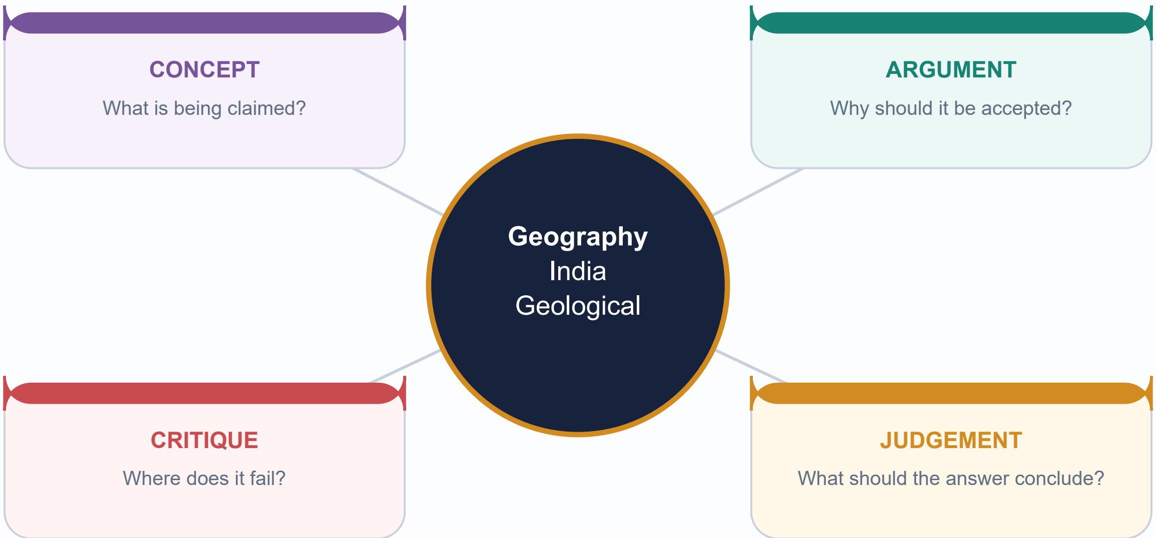
Geography 02 - India Geological Structure - Solved Practice Workbook

GS-I + Prelims | 2026-08-15

Direct PYQs plus clearly labelled adjacent applications

Strict original-MCQ rotation A -> B -> C -> D

2026 Prelims answer remains provisional, not officially final.



Facts from official sources | Inferences clearly marked | No fabricated data

HIGH UPSC RELEVANCE

HIGH

PYQ 2022 GS-I Q4 - Characteristics and types of primary rocks (10 marks)

GS-I / Prelims
Solved verified PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02.

STATIC THEORY

- Demand: Describe the characteristics and types of primary rocks.

- Solution: [CLAIM] Primary rocks are igneous rocks formed by solidification of magma or lava and constitute the original mineral source for later sedimentary and metamorphic rocks. [EVIDENCE] Intrusive or plutonic rocks such as granite and gabbro cool slowly at depth and therefore commonly develop coarse crystals. [MECHANISM] Insulation delays cooling and permits crystal growth. [EVIDENCE] Extrusive rocks such as basalt and rhyolite cool rapidly at or near the surface and are commonly fine-grained; glassy or vesicular textures may occur when cooling is very rapid or gases escape. [MECHANISM] Composition ranges broadly from silica-rich felsic to iron-magnesium-rich mafic types, influencing colour, density and weathering. [EVIDENCE] Indian examples include the Deccan flood basalts and widespread granitic-gneissic terrains of the Peninsular shield. [LIMIT] 'Primary' describes genetic origin, not present exposure or economic value; later alteration can metamorphose an igneous protolith. [VERDICT] Thus texture, composition and mode of occurrence together classify primary rocks and explain their landscape and resource significance.

- Why this earns marks: it follows the directive 'describe', classifies both occurrence and composition, uses Indian evidence, gives mechanism and avoids confusing primary with oldest.

MUST-KNOW FACTS

1. Identify ownership.

2. Decode the directive.

3. Use named evidence and qualification.

UPSC TRAPS

WRONG: Present a provisional key as final.

CORRECT: The 2026 key remains explicitly provisional.

MAINS ANGLE

Why this earns marks: it follows the directive 'describe', classifies both occurrence and composition, uses Indian evidence, gives mechanism and avoids confusing primary with oldest.

STUDY LINK

Local official paper/key text and routing ledger.

HIGH

PYQ 2025 GS-I Q16 - Continents and ocean basins due to crustal tectonics (15 marks)

GS-I / Prelims
Solved verified PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02; applied here to India's plate history.

STATIC THEORY

- Demand: Discuss the changes in the continents and ocean basins due to crustal tectonics.

- Solution: [CLAIM] Continents and ocean basins are temporary configurations within a plate cycle rather than permanent containers. [EVIDENCE] Matching continental margins, continuous ancient rock belts, fossil distributions, palaeoclimate and palaeomagnetism establish former continental connections. [MECHANISM] At divergent boundaries, rifting and sea-floor spreading create new oceanic crust, progressing from continental rift to narrow sea and mature ocean. At convergent boundaries, subduction consumes oceanic lithosphere, forms trenches and arcs, and progressively closes basins. Continent-continent collision then shortens and thickens buoyant crust, producing suture zones and fold-thrust mountains. [EVIDENCE] The East African Rift, Red Sea, Atlantic, Pacific, Mediterranean and Himalayan-Tethyan suture illustrate successive states. India's separation from Gondwana, northward motion and collision with Eurasia demonstrate continental translation, Tethys closure, Himalayan uplift and foreland-basin creation. [ANALYSIS] Transform motion offsets ridges and margins without creating or destroying crust, while sedimentation reshapes passive margins and deltas. [LIMIT] The Wilson-cycle sequence is conceptual: rates vary, microplates and terranes complicate margins, and not every basin completes the full cycle. [VERDICT] Crustal tectonics continuously redistributes land and sea while organising global relief, resources and geophysical hazards.

- Why this earns marks: it answers both continents and oceans, integrates evidence with mechanism, uses India and world examples, and qualifies the idealised cycle.

MUST-KNOW FACTS

1. Identify ownership.

2. Decode the directive.

3. Use named evidence and qualification.

UPSC TRAPS

WRONG: Present a provisional key as final.

CORRECT: The 2026 key remains explicitly provisional.

MAINS ANGLE

Why this earns marks: it answers both continents and oceans, integrates evidence with mechanism, uses India and world examples, and qualifies the idealised cycle.

STUDY LINK

Local official paper/key text and routing ledger.

HIGH

PYQ 2024 Prelims Q10 - Fold/block mountain matching

GS-I / Prelims
Solved verified PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02. Official Set-A key: B.

STATIC THEORY

- Demand: Rows tested Vosges as fold, Alps as block, Appalachians as fold and Andes as fold; asked how many were correctly matched.

- Solution: [FACT] Vosges is a block mountain, so row 1 is incorrect. Alps is a fold mountain, so row 2 is incorrect. Appalachians and Andes are fold-mountain systems, so rows 3 and 4 are correct. Therefore two rows are correct: option B.

- Elimination hinge: reverse the first two classifications; do not confuse a rift-flank horst with a young fold belt.

MUST-KNOW FACTS

1. Identify ownership.

2. Decode the directive.

3. Use named evidence and qualification.

UPSC TRAPS

WRONG: Present a provisional key as final.

CORRECT: The 2026 key remains explicitly provisional.

MAINS ANGLE

Elimination hinge: reverse the first two classifications; do not confuse a rift-flank horst with a young fold belt.

STUDY LINK

Local official paper/key text and routing ledger.

HIGH

PYQ 2025 Prelims Q25 - Evidences of continental drift

GS-I / Prelims
Solved verified PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02. Official Set-A key: C.

STATIC THEORY

- Demand: Statements covered matching ancient rocks of Brazil and western Africa, Ghana gold derived from the Brazil plateau when joined, and Gondwana sediment counterparts across southern landmasses.

- Solution: [FACT] All three were accepted as evidence in the official key: geological continuity, placer/mineral-source reconstruction and cross-Gondwana stratigraphic correlation. Therefore option C.

- Elimination hinge: continental-drift evidence is broader than coastline fit; it includes lithologic, mineral, fossil, climatic and stratigraphic continuity.

MUST-KNOW FACTS

1. Identify ownership.

2. Decode the directive.

3. Use named evidence and qualification.

UPSC TRAPS

WRONG: Present a provisional key as final.

CORRECT: The 2026 key remains explicitly provisional.

MAINS ANGLE

Elimination hinge: continental-drift evidence is broader than coastline fit; it includes lithologic, mineral, fossil, climatic and stratigraphic continuity.

STUDY LINK

Local official paper/key text and routing ledger.

HIGH

PYQ 2026 Prelims Q34 - Peninsular Block characteristics

GS-I / Prelims
Solved verified PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02. PROVISIONAL ANSWER - NOT OFFICIALLY FINAL.

STATIC THEORY

- Demand: Statements: western-coast submergence due to tectonic activity; residual ranges such as Veliconda and Mahendragiri; deep V-shaped valleys formed by fast-flowing rivers.

- Solution: [FACT] Statements 1 and 2 match standard descriptions of the Peninsular Block: parts of the west coast show submergence/faulted-margin expression, and residual hill ranges occur within the ancient denuded terrain. [ANALYSIS] Statement 3 is not a general Peninsular-block characteristic; mature, structurally controlled valleys are more typical than youthful deep V-shaped valleys as a defining trait. The provisional Set-A key records option B (1 and 2). [LIMIT] The 2026 key is provisional at the repository cutoff and must not be presented as a final UPSC key.

- Elimination hinge: distinguish isolated youthful incision from the broad geomorphic character of an old denuded shield.

MUST-KNOW FACTS

1. Identify ownership.

2. Decode the directive.

3. Use named evidence and qualification.

UPSC TRAPS

WRONG: Present a provisional key as final.

CORRECT: The 2026 key remains explicitly provisional.

MAINS ANGLE

Elimination hinge: distinguish isolated youthful incision from the broad geomorphic character of an old denuded shield.

STUDY LINK

Local official paper/key text and routing ledger.

HIGH

Adjacent PYQ 2022 GS-I Q6 - Natural-resource potential of the Deccan Trap

GS-I / Prelims
Solved verified PYQ

NEWS TRIGGER

True owner: Geography Topic 31; included as a bounded geology application.

STATIC THEORY

- Demand: Discuss the natural resource potentials of the Deccan Trap.

- Solution: [CLAIM] The Deccan Trap's resource potential arises from its layered basalt architecture and later weathering, not from basalt alone. [EVIDENCE] Weathered basalt contributes parent material to extensive black-soil tracts supporting cotton, oilseeds and other crops. [MECHANISM] Clay-rich weathering products retain moisture, though drainage and climate modify productivity. [EVIDENCE] Vesicular tops, fractures, weathered contacts and intertrappean beds form local aquifers; compact flow interiors may be poor aquifers. [EVIDENCE] Basalt is used as construction aggregate and building stone, while lateritic caps and associated weathering profiles can host local industrial-mineral potential. [ANALYSIS] Step relief, plateaux and escarpments create tourism and hydropower/transport opportunities in selected sectors. [LIMIT] Groundwater is heterogeneous, black soil can face waterlogging or cracking, and not all critical minerals are hosted by Deccan basalt. [VERDICT] The province is best understood as a soil-water-construction-landscape resource system whose benefits require location-specific management.

- Why this earns marks: it gives several resource dimensions, derives each from flow/weathering structure and avoids an exaggerated mineral list.

MUST-KNOW FACTS

1. Identify ownership.
2. Decode the directive.
3. Use named evidence and qualification.

UPSC TRAPS

WRONG: Present a provisional key as final.

CORRECT: The 2026 key remains explicitly provisional.

MAINS ANGLE

Why this earns marks: it gives several resource dimensions, derives each from flow/weathering structure and avoids an exaggerated mineral list.

STUDY LINK

Local official paper/key text and routing ledger.

HIGH

Adjacent PYQ 2021 GS-I Q5 - Gondwanaland mineral base and mining share

GS-I / Prelims
Solved verified PYQ

NEWS TRIGGER

True owner: Geography Topic 31; geology application included without duplicating mining economics.

STATIC THEORY

- Demand: Discuss why India is well endowed with mineral resources due to Gondwanaland yet mining contributes relatively little to GDP.

- Solution: [CLAIM] Gondwana-derived shield and basin history endowed India with diverse metallic ores and coal, but geological endowment is not identical to realised economic value. [EVIDENCE] Ancient Peninsular mobile belts host iron, manganese and base-metal provinces, while faulted Gondwana basins host major coalfields. [MECHANISM] Magmatism, metamorphism, hydrothermal activity, sedimentation and long preservation created deposits. [ANALYSIS] Low value addition, variable ore grade, deep or fragmented deposits, infrastructure gaps, environmental-social constraints, exploration uncertainty and cyclical prices can restrict mining's GDP contribution. [LIMIT] Gondwanaland should not be used as a blanket explanation for every mineral or petroleum basin, and current GDP shares require dated official data. [VERDICT] Policy must convert geological knowledge into responsible exploration, beneficiation, recycling and community-sensitive value chains rather than equating extraction volume with development.

- Why this earns marks: it separates endowment from economic conversion, uses geological mechanisms and preserves the true Topic 31 ownership boundary.

MUST-KNOW FACTS

1. Identify ownership.
2. Decode the directive.
3. Use named evidence and qualification.

UPSC TRAPS

WRONG: Present a provisional key as final.

CORRECT: The 2026 key remains explicitly provisional.

MAINS ANGLE

Why this earns marks: it separates endowment from economic conversion, uses geological mechanisms and preserves the true Topic 31 ownership boundary.

STUDY LINK

Local official paper/key text and routing ledger.

HIGH

PYQ 2018 Prelims Q57 - Magnetic reversal and early atmosphere

Prelims Solved verified-demand PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02. INFERRED ANSWER - NOT OFFICIALLY VERIFIED LOCALLY.

STATIC THEORY

- Question/demand: Statements tested whether Earth's magnetic field has reversed over geological time, whether the earliest atmosphere had 54 per cent oxygen and no carbon dioxide, and whether living organisms modified the early atmosphere.

- [FACT] Statement 1 is correct in concept: palaeomagnetic records preserve repeated polarity reversals, although reversal intervals are irregular and 'every few hundred thousand years' is only a broad formulation.

- [FACT] Statement 2 is false: the early Earth did not possess an oxygen-rich atmosphere of the stated composition; substantial free oxygen accumulated much later.

- [FACT] Statement 3 is correct: photosynthetic life profoundly altered atmospheric chemistry by producing oxygen and participating in long-term carbon cycling.

- INFERRED ANSWER: statements 1 and 3 only, option C. The local repository does not hold a readable official 2018 key.

MUST-KNOW FACTS

1. Magnetic reversals are recorded by remanent magnetism in suitable rocks.

2. The early atmosphere was not 54 per cent oxygen.

3. Life, especially photosynthesis, modified atmospheric composition.

UPSC TRAPS

WRONG: Present option C as locally official.

CORRECT: Label it inferred because the local official key is unavailable.

WRONG: Treat reversal spacing as perfectly periodic.

CORRECT: Reversal intervals are irregular.

MAINS ANGLE

Use palaeomagnetism as evidence for geologic change and plate reconstruction.

STUDY LINK

Core Topic 02 evidence and dating.

HIGH

PYQ 2023 Prelims Q9 - Amarkantak, Biligirirangan and Seshachalam

Prelims Solved locally held PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02. INFERRED ANSWER - NOT OFFICIALLY VERIFIED LOCALLY.

STATIC THEORY

- Question/demand: Statements claimed Amarkantak lies at the confluence of Vindhya and Sahyadri, Biligirirangan is the easternmost Satpura, and Seshachalam is the southernmost Western Ghats.

- [FACT] Statement 1 is incorrect: Amarkantak is associated with the Vindhya-Satpura meeting region, not the Sahyadri/Western Ghats.

- [FACT] Statement 2 is incorrect: Biligirirangan lies in southern Karnataka at an Eastern Ghats-Western Ghats biogeographic linkage, not the eastern Satpura.

- [FACT] Statement 3 is incorrect: Seshachalam is part of the Eastern Ghats in Andhra Pradesh, not the southern Western Ghats.

- INFERRED ANSWER: none of the statements is correct, option D. The local question text is held, but the routing ledger records the official key as unavailable.

MUST-KNOW FACTS

1. Amarkantak: Vindhya-Satpura highland context.

2. Biligirirangan: southern Karnataka, Eastern-Western Ghats linkage.

3. Seshachalam: Eastern Ghats, Andhra Pradesh.

UPSC TRAPS

WRONG: Attach a true hill name to a plausible but wrong parent range.

CORRECT: Locate each hill independently on a mental map.

WRONG: Present option D as locally official.

CORRECT: Keep the inferred-key label.

MAINS ANGLE

Map-based elimination tests geological and physiographic regional placement.

STUDY LINK

Peninsular shield map and regional-comparison practice.

HIGH

Concept and Map MCQs 1-4

Prelims Solved practice

NEWS TRIGGER

Strict key rotation A -> B -> C -> D.

STATIC THEORY

- Q1. Which statement best distinguishes geological structure from physiography? A) Structure concerns rock architecture; physiography concerns broad surface divisions B) Both terms mean relief only C) Physiography concerns mineral chemistry only D) Structure concerns climate zones

- Q2. Which is the safest description of the Peninsular block? A) Uniform Archaean slab B) Composite ancient shield cut by later rifts and partly covered by younger rocks C) Young volcanic arc D) Sediment-filled foredeep

- Q3. Which evidence directly helps reconstruct past plate latitude? A) Modern rainfall B) River discharge C) Palaeomagnetism D) Population density

- Q4. Which statement about radiometric dating is correct? A) It always dates sediment deposition B) It makes stratigraphy unnecessary C) It gives the same age for all minerals D) It dates a mineral event that needs geological interpretation

- DETAILED SOLUTIONS

- Q1 - A. Structure is the hidden rock/tectonic framework; physiography is its broad surface expression.

- Q2 - B. The Peninsula is a mosaic of cratons/mobile belts with later basins, faults and lava cover.

- Q3 - C. Magnetic remanence in rocks records ancient field orientation after careful correction.

- Q4 - D. The measured isotopic clock may represent crystallisation, metamorphism or cooling.

MUST-KNOW FACTS

1. Cover nearly every subtopic.
2. Use elimination before recall.
3. Track repeated conceptual errors.

UPSC TRAPS

WRONG: Read the key before attempting.

CORRECT: Attempt and justify every rejected option.

MAINS ANGLE

Each explanation is a compact evidence unit for GS-I.

STUDY LINK

Return to the matching main-PDF section for remediation.

HIGH

Concept and Map MCQs 5-8

Prelims Solved practice

NEWS TRIGGER

Strict key rotation A -> B -> C -> D.

STATIC THEORY

- Q5. Gondwana in 'Gondwana Supergroup' refers primarily to A) a continental sedimentary succession in faulted basins B) the present Indian plate boundary C) the Deccan lava province D) the Himalayan crystalline core

- Q6. The main resource linkage of classic Gondwana basins is A) offshore petroleum only B) major coal measures in continental rift basins C) coral limestone D) gold-bearing greenstone belts

- Q7. Which pair is correctly matched? A) Vindhyan-flood basalt B) Dharwar-modern alluvium C) Deccan Trap-stacked basalt flows D) Siwalik-oldest craton

- Q8. Which is NOT sufficient to infer petroleum? A) sedimentary basin B) source rock C) reservoir and seal D) basin presence alone

- DETAILED SOLUTIONS

- Q5 - A. The stratigraphic succession is distinct from Gondwana the palaeocontinent.

- Q6 - B. Coal-bearing sediment cycles accumulated in subsiding continental basins.

- Q7 - C. Deccan Trap is an extrusive flood-basalt province.

- Q8 - D. A complete petroleum system and timing are required.

MUST-KNOW FACTS

1. Cover nearly every subtopic.

2. Use elimination before recall.

3. Track repeated conceptual errors.

UPSC TRAPS

WRONG: Read the key before attempting.

CORRECT: Attempt and justify every rejected option.

MAINS ANGLE

Each explanation is a compact evidence unit for GS-I.

STUDY LINK

Return to the matching main-PDF section for remediation.

HIGH

Concept and Map MCQs 9-12

Prelims Solved practice

NEWS TRIGGER

Strict key rotation A -> B -> C -> D.

STATIC THEORY

- Q9. Siwalik rocks are best described as A) young foreland molasse folded at the Himalayan front B) oldest Himalayan crystalline basement C) oceanic crust of the Tethys D) Deccan basalt outliers

- Q10. Which order is broadly south-to-north? A) Greater-Lesser-Siwalik-plain B) Siwalik-Lesser-Greater-Tethyan/Trans-Himalaya C) Tethys-Siwalik-plain-Greater D) Plain-Trans-Himalaya-Siwalik-Lesser

- Q11. The Indo-Ganga-Brahmaputra Plain is structurally A) an exposed craton B) a volcanic plateau C) a sediment-filled foreland basin D) an island arc

- Q12. Which distinction is correct? A) Both island groups are coral B) Lakshadweep is an active arc C) Andaman is a shield fragment D) Andaman-Nicobar is arc-related; Lakshadweep is coral-atoll terrain

- DETAILED SOLUTIONS

- Q9 - A. They are erosion products of the rising Himalaya deposited in a foreland setting.

- Q10 - B. The simplified cross-section runs outer foothills to higher crystalline and northern domains.

- Q11 - C. Flexure under Himalayan loading created accommodation filled by alluvium.

- Q12 - D. Their geological settings and hazards differ fundamentally.

MUST-KNOW FACTS

1. Cover nearly every subtopic.

2. Use elimination before recall.

3. Track repeated conceptual errors.

UPSC TRAPS

WRONG: Read the key before attempting.

CORRECT: Attempt and justify every rejected option.

MAINS ANGLE

Each explanation is a compact evidence unit for GS-I.

STUDY LINK

Return to the matching main-PDF section for remediation.

HIGH

Concept and Map MCQs 13-16

Prelims Solved practice

NEWS TRIGGER

Strict key rotation A -> B -> C -> D.

STATIC THEORY

- Q13. The Narmada and Tapi are high-yield examples of A) structurally controlled west-flowing trough rivers B) glacial antecedent rivers C) delta-building east-flowing rivers D) coral-island streams

- Q14. Groundwater in crystalline hard rock commonly depends on A) abundant primary pore space B) weathering, joints and fractures C) coal seams only D) marine fossils

- Q15. Which statement best controls resource determinism? A) Every old rock is an ore B) Every basin has petroleum C) Geology creates potential but grade, system completeness and economics decide resources D) Minerals follow state boundaries

- Q16. Which is the correct seismic caution? A) Peninsula is aseismic B) Zone V predicts the next earthquake C) Alluvium always protects buildings D) Relative stability does not exclude damaging intraplate earthquakes

- DETAILED SOLUTIONS

- Q13 - A. They occupy major structural troughs across the regional slope.

- Q14 - B. Secondary permeability controls most hard-rock aquifers.

- Q15 - C. Geological association is necessary in many cases but not economically sufficient.

- Q16 - D. Kachchh, Latur, Koyna and Jabalpur show inherited structures can reactivate.

MUST-KNOW FACTS

1. Cover nearly every subtopic.

2. Use elimination before recall.

3. Track repeated conceptual errors.

UPSC TRAPS

WRONG: Read the key before attempting.

CORRECT: Attempt and justify every rejected option.

MAINS ANGLE

Each explanation is a compact evidence unit for GS-I.

STUDY LINK

Return to the matching main-PDF section for remediation.

HIGH

Concept and Map MCQs 17-20

Prelims Solved practice

NEWS TRIGGER

Strict key rotation A -> B -> C -> D.

STATIC THEORY

- Q17. Which unit is dominantly a Proterozoic sedimentary basin succession? A) Cuddapah B) Deccan Trap C) Dharwar greenstone belt D) Himalayan Frontal Thrust

- Q18. Panna is the classic location cue for A) Gondwana coal B) Vindhyan diamond association C) Deccan lava D) Siwalik petroleum

- Q19. Which feature most strongly distinguishes a craton? A) Recent volcanic arc B) Unconsolidated delta C) Long-lived ancient continental nucleus D) Active ocean ridge

- Q20. A mapped lineament should be treated as A) proof of active faulting B) a seismic prediction C) a mineral reserve D) a linear feature requiring geological validation

- DETAILED SOLUTIONS

- Q17 - A. Cuddapah is a thick basin fill over older basement.

- Q18 - B. Use it as a bounded association, not a universal Vindhyan property.

- Q19 - C. Cratons preserve old, comparatively stable continental crust.

- Q20 - D. It may reflect several kinds of geological or geomorphic boundaries.

MUST-KNOW FACTS

1. Cover nearly every subtopic.
2. Use elimination before recall.
3. Track repeated conceptual errors.

UPSC TRAPS

WRONG: Read the key before attempting.
CORRECT: Attempt and justify every rejected option.

MAINS ANGLE

Each explanation is a compact evidence unit for GS-I.

STUDY LINK

Return to the matching main-PDF section for remediation.

HIGH

Concept and Map MCQs 21-24

Prelims Solved practice

NEWS TRIGGER

Strict key rotation A -> B -> C -> D.

STATIC THEORY

- Q21. Which is a correct Deccan aquifer statement? A) Interflow weathered/vesicular zones can store water B) Every compact basalt flow is highly porous C) Groundwater occurs only in rivers D) Dykes always increase permeability

- Q22. Which basin is most directly linked to a north-south rift in Gujarat? A) Damodar B) Cambay C) Spiti D) Bastar

- Q23. Which process creates the Himalayan foreland basin? A) Coral growth B) Mid-ocean spreading C) Flexure under orogenic load D) Aeolian deflation

- Q24. Which is the safest statement on collision chronology? A) One exact date applies everywhere B) Collision ended tectonic activity C) No Tethyan ocean existed D) Collision was progressive and definitions/dates vary

- DETAILED SOLUTIONS

- Q21 - A. Basalt aquifers are heterogeneous and structurally controlled.

- Q22 - B. Cambay is a rifted sedimentary basin.

- Q23 - C. Mountain loading bends the Indian plate and creates a foredeep.

- Q24 - D. Different evidence records different stages of convergence and collision.

MUST-KNOW FACTS

1. Cover nearly every subtopic.

2. Use elimination before recall.

3. Track repeated conceptual errors.

UPSC TRAPS

WRONG: Read the key before attempting.

CORRECT: Attempt and justify every rejected option.

MAINS ANGLE

Each explanation is a compact evidence unit for GS-I.

STUDY LINK

Return to the matching main-PDF section for remediation.

HIGH

Concept and Map MCQs 25-28

Prelims Solved practice

NEWS TRIGGER

Strict key rotation A -> B -> C -> D.

STATIC THEORY

- Q25. Which pair is correctly matched? A) Dharwar-ancient supracrustal/metamorphic belts B) Gondwana-flood basalt C) Vindhyan-island arc D) Siwalik-craton

- Q26. The best evidence that an ocean floor is spreading is A) oldest crust at ridge B) symmetrical magnetic stripes and increasing age away from ridge C) coal on continents D) folding in a delta

- Q27. Which setting best explains Barren Island? A) coral atoll B) stable shield C) subduction-related island arc volcanism D) foreland molasse

- Q28. Which is NOT a valid reason to call the Peninsula stable? A) lower average deformation than Himalaya B) ancient cratonic crust C) long denudation D) complete absence of faults and earthquakes

- DETAILED SOLUTIONS

- Q25 - A. Dharwar usage is tied to ancient shield belts.

- Q26 - B. Magnetic symmetry and age pattern directly record creation and outward movement.

- Q27 - C. It belongs to the Andaman arc setting.

- Q28 - D. Stable is relative, not absolute.

MUST-KNOW FACTS

1. Cover nearly every subtopic.

2. Use elimination before recall.

3. Track repeated conceptual errors.

UPSC TRAPS

WRONG: Read the key before attempting.

CORRECT: Attempt and justify every rejected option.

MAINS ANGLE

Each explanation is a compact evidence unit for GS-I.

STUDY LINK

Return to the matching main-PDF section for remediation.

HIGH

Concept and Map MCQs 29-32

Prelims Solved practice

NEWS TRIGGER

Strict key rotation A -> B -> C -> D.

STATIC THEORY

- Q29. Which statement on Cuddapah/Vindhyan is correct? A) Both rest on older basement with major unconformities B) Both are Cenozoic deltas C) Both are basalt provinces D) Both are identical in every source

- Q30. Coal formation in Gondwana basins required A) only faulting B) organic accumulation, burial and preservation in subsiding basins C) coral reefs D) granite crystallisation

- Q31. Which is a correct Himalaya statement? A) One simple fold B) Entirely Tertiary sediment C) Imbricated fold-thrust belt with multiple rock units D) Volcanic arc like Andes

- Q32. Which statement on BIS zoning is safest at the 2026 cutoff? A) Zone I is highest B) Zone VI is official C) Zoning predicts events D) Public classification uses Zones II-V

- DETAILED SOLUTIONS

- Q29 - A. They are separate Proterozoic basin successions.

- Q30 - B. Faulting created space, but depositional and burial conditions made coal.

- Q31 - C. The belt contains thrust sheets, folds and crystalline/metamorphic domains.

- Q32 - D. Use current official public classification and distinguish hazard from prediction.

MUST-KNOW FACTS

1. Cover nearly every subtopic.
2. Use elimination before recall.
3. Track repeated conceptual errors.

UPSC TRAPS

WRONG: Read the key before attempting.

CORRECT: Attempt and justify every rejected option.

MAINS ANGLE

Each explanation is a compact evidence unit for GS-I.

STUDY LINK

Return to the matching main-PDF section for remediation.

HIGH

Concept and Map MCQs 33-36

Prelims Solved practice

NEWS TRIGGER

Strict key rotation A -> B -> C -> D.

STATIC THEORY

- Q33. Which is the best example of lithologic control on relief? A) Resistant band forming a scarp B) Longitude producing coal C) Population causing gneiss D) Monsoon creating a fault

- Q34. Which unit is an outer Himalayan foreland deposit? A) Archaean gneiss B) Siwalik molasse C) Deccan Trap D) Dharwar schist

- Q35. Which evidence source best images buried basin geometry? A) political map B) rain gauge C) seismic reflection D) crop calendar

- Q36. Which conclusion is most defensible? A) Geology determines society B) Minerals occur uniformly by age C) All plains are shields D) Geology organises opportunities and hazards but other controls mediate outcomes

- DETAILED SOLUTIONS

- Q33 - A. Differential resistance affects erosion and relief.

- Q34 - B. Siwalik sediments were derived from the rising orogen.

- Q35 - C. Seismic methods reveal subsurface layer and fault geometry.

- Q36 - D. This is the required anti-determinist qualification.

MUST-KNOW FACTS

1. Cover nearly every subtopic.
2. Use elimination before recall.
3. Track repeated conceptual errors.

UPSC TRAPS

WRONG: Read the key before attempting.
CORRECT: Attempt and justify every rejected option.

MAINS ANGLE

Each explanation is a compact evidence unit for GS-I.

STUDY LINK

Return to the matching main-PDF section for remediation.

HIGH

Concept and Map MCQs 37-40

Prelims Solved practice

NEWS TRIGGER

Strict key rotation A -> B -> C -> D.

STATIC THEORY

- Q37. Which map set most directly answers India geological structure? A) Peninsula-Himalaya-foreland provinces B) language families C) cropping intensity D) urban hierarchy

- Q38. The term 'trap' in Deccan Trap refers to A) a petroleum trap B) step-like topography of lava flows C) a coal mine D) an ocean trench

- Q39. Which process can concentrate coastal heavy minerals? A) crustal thickening alone B) radiometric decay C) wave/current sorting of resistant dense grains D) glacial plucking in Lakshadweep

- Q40. Which is the correct answer-writing sequence? A) facts without thesis B) map without explanation C) one resource list D) claim-evidence-mechanism-qualification-verdict

- DETAILED SOLUTIONS

- Q37 - A. The three-province map supplies the national structural frame.

- Q38 - B. The stacked flows create stair-like relief.

- Q39 - C. Erosion, transport and hydraulic sorting form placer concentrations.

- Q40 - D. This sequence converts recall into analysis.

MUST-KNOW FACTS

1. Cover nearly every subtopic.
2. Use elimination before recall.
3. Track repeated conceptual errors.

UPSC TRAPS

WRONG: Read the key before attempting.
CORRECT: Attempt and justify every rejected option.

MAINS ANGLE

Each explanation is a compact evidence unit for GS-I.

STUDY LINK

Return to the matching main-PDF section for remediation.

HIGH

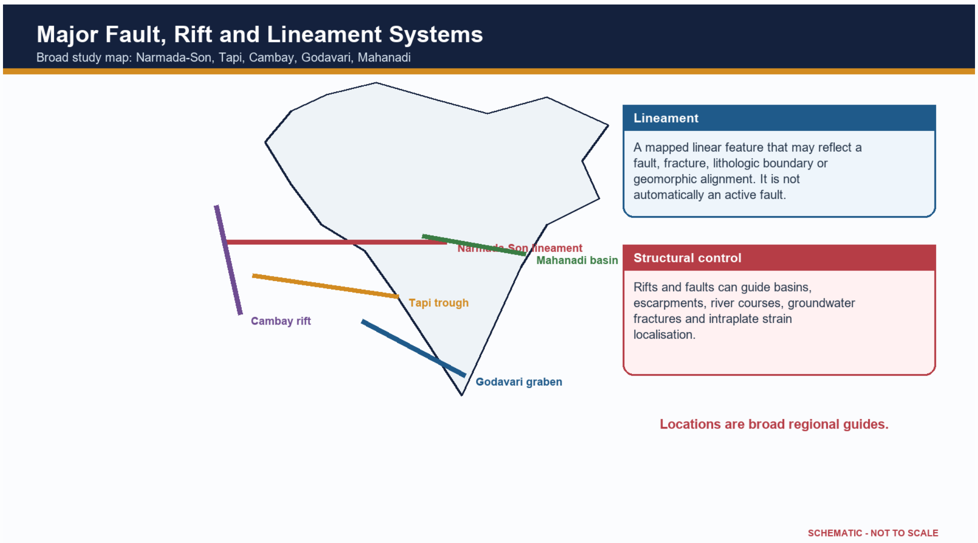
Map and cross-section workbook

GS-I / Prelims
Spatial practice

NEWS TRIGGER

Draw every figure schematic/not to scale.

FAULT/RIFT STUDY MAP



Broad zones only.
Original schematic.

STATIC THEORY

- Exercise 1: On a blank India outline mark the three structural provinces, Deccan Trap, Dharwar, Singhbhum, Cuddapah, Vindhyan, Damodar-Godavari basins, Narmada-Son, Cambay, Andaman-Nicobar and Lakshadweep.

- Solution: keep Himalayan arc north, foreland immediately south, ancient shield across the Peninsula, basalt west-central, Gondwana troughs east/south-central, island arc east and coral atolls west.

- Exercise 2: Draw the Himalayan cross-section with foreland, Siwalik, MBT, Lesser, MCT, Greater, Tethyan/Trans-Himalayan and suture region.

- Solution caution: thrust order is simplified; use HFT/MFT nomenclature consistently and never label Siwalik oldest.

- Exercise 3: Draw a Gondwana graben with normal faults, subsidence, sediment cycles and coal seams; add the statement 'not every rift is coal-bearing'.

- Exercise 4: Draw Deccan stacked flows showing compact core, vesicular/weathered top, intertrappean bed, dyke/fracture and variable groundwater role.

MUST-KNOW FACTS

1. Map placement must support the written mechanism.

2. Cross-sections need direction arrows.

3. Never imply exact legal boundaries.

UPSC TRAPS

WRONG: Decorative unlabelled India outline.

CORRECT: Use 5-8 relevant labels and one causal caption.

MAINS ANGLE

A 30-second cross-section can replace a long descriptive paragraph.

STUDY LINK

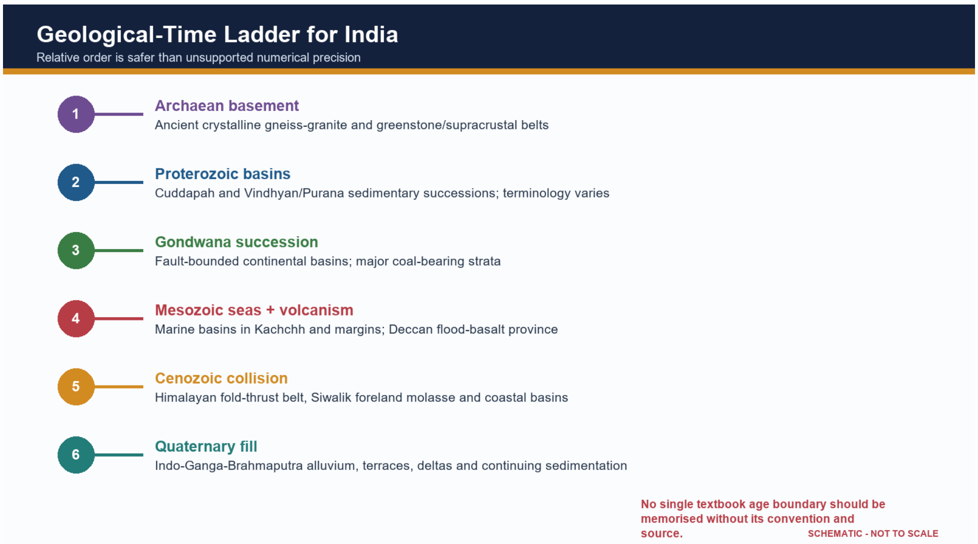
Main visuals 4-10 and 16.

HIGH

Chronology, hierarchy and remedial exercises

Prelims / GS-I
Remedial practice

RELATIVE-TIME LADDER



Relative sequence; boundaries vary.

Original schematic.

STATIC THEORY

- Order: Archaean-Dharwar -> Cuddapah/Vindhyan -> Gondwana -> Deccan -> Himalayan collision/Cenozoic basins -> Quaternary alluvium.

- Differentiate: Gondwana palaeocontinent versus Gondwana Supergroup; craton versus shield; lineament versus active fault; stratification versus metamorphic foliation.

- Correct the errors: 'Deccan Trap is sedimentary'; 'Siwalik is oldest'; 'Northern Plain is shield'; 'Peninsula is aseismic'; 'all islands are coral'; 'every basin has petroleum'.

- Remedial method: write the correct mechanism beside each false statement and add one named Indian region.

MUST-KNOW FACTS

1. Relative order earns safer marks than invented numerical precision.

2. Terminology must carry its source convention.

UPSC TRAPS

WRONG: Memorise old fourfold schemes without warning.

CORRECT: Use them only with an explicit older-textbook convention.

MAINS ANGLE

Turn each corrected error into a one-line conclusion qualifier.

STUDY LINK

Main sections 2, 5-12.

HIGH

10 marks: Distinguish India's geological structure from its physiographic divisions. Illustrate with two regions.

GS-I Solved Mains workbook

NEWS TRIGGER

Examiner-grade model with evidence scaled to marks.

STATIC THEORY

- [CLAIM] Geological structure is the rock-and-tectonic architecture beneath India, whereas physiography is its broad surface expression. [EVIDENCE] The Peninsular Plateau is a physiographic unit, but geologically it contains Dharwar, Bastar, Singhbhum and Bundelkhand blocks, Proterozoic basins, Gondwana rifts and Deccan basalt. [MECHANISM] Differential resistance, faulting and long denudation generate residual hills, scarps and structurally guided rivers. [EVIDENCE] The Northern Plain appears as level physiography but geologically is a foreland depression filled by Himalayan and peninsular alluvium. [MECHANISM] Orogenic loading caused flexure and sediment created the plain. [LIMIT] Climate and rivers continue to reshape both provinces; structure is a template, not the sole cause. [VERDICT] Thus physiography is best read as the visible, modified expression of deeper geology. Why this earns marks: it defines both terms, uses two named regions, explains mechanisms and qualifies determinism.

MUST-KNOW FACTS

1. Claim first.

2. Named region second.

3. Mechanism third.

4. Qualification and verdict last.

UPSC TRAPS

WRONG: Use the same evidence volume for every mark level.

CORRECT: Scale dimensions and examples to 10/15/20 marks.

MAINS ANGLE

it defines both terms, uses two named regions, explains mechanisms and qualifies determinism.

STUDY LINK

Visual 16 answer spine.

HIGH

15 marks: Explain how the geological evolution of Peninsular India controls coal, metallic minerals and groundwater.

GS-I Solved
Mains workbook

NEWS TRIGGER

Examiner-grade model with evidence scaled to marks.

STATIC THEORY

- [CLAIM] Peninsular resource geography reflects distinct stages of crustal evolution rather than one uniformly mineral-rich shield. [EVIDENCE] Ancient Dharwar-Singhbhum-Aravalli mobile belts experienced magmatism, metamorphism and hydrothermal activity, creating iron, manganese, gold and base-metal provinces. [MECHANISM] Ore localisation followed specific mineralising events and structural traps. [EVIDENCE] Later faulting formed Damodar, Son-Mahanadi and Godavari Gondwana basins where continental sediments and organic matter accumulated. [MECHANISM] Subsidence, burial and preservation produced major coal measures. [EVIDENCE] Crystalline rocks have low primary porosity, but weathered mantles, joints and faults store groundwater; Deccan basalt adds vesicular tops, fractures and interflow contacts. [ANALYSIS] Therefore metallic ore, coal and aquifer maps differ because their geological mechanisms differ. [LIMIT] Grade, recharge, technology, environmental constraints and economics decide usable resources. [VERDICT] The Peninsula is best understood as a layered resource system: ancient mineralised crust, younger sedimentary basins and structurally controlled aquifers. Why this earns marks: it covers all three demands with named belts, distinct mechanisms and an economic-hydrological qualification.

MUST-KNOW FACTS

1. Claim first.
2. Named region second.
3. Mechanism third.
4. Qualification and verdict last.

UPSC TRAPS

WRONG: Use the same evidence volume for every mark level.

CORRECT: Scale dimensions and examples to 10/15/20 marks.

MAINS ANGLE

it covers all three demands with named belts, distinct mechanisms and an economic-hydrological qualification.

STUDY LINK

Visual 16 answer spine.

HIGH

20 marks: 'India's geological stability is relative, spatially uneven and socially consequential.' Analyse.

GS-I Solved
Mains workbook

NEWS TRIGGER

Examiner-grade model with evidence scaled to marks.

STATIC THEORY

- [CLAIM] India's geological stability is a comparison of deformation regimes, not a division between active and inactive territory. [EVIDENCE] The Himalaya and north-east occupy an active collision system with thrusting, crustal shortening and frequent earthquakes. [MECHANISM] Continuing India-Eurasia convergence stores and releases elastic strain; adjoining alluvial basins can amplify shaking. [EVIDENCE] Andaman-Nicobar lies on a subduction-related arc with earthquake, tsunami and volcanic contexts. [EVIDENCE] Peninsular cratons are older and less deforming on average, yet inherited rifts and faults produce intraplate hazards in Kachchh and events such as Koyna, Latur and Jabalpur. [MECHANISM] Far-field stress and local structural weaknesses can reactivate old zones. [ANALYSIS] Social consequence depends on exposure: dense Himalayan towns, vulnerable masonry, unstable slopes, dams, industrial corridors and soft-sediment cities transform geological processes into risk. Resource consequences are also uneven: shield belts host metallic ores, Gondwana rifts coal, and sedimentary margins petroleum systems. [LIMIT] Seismic zonation does not predict earthquakes, and resource association does not ensure economic extraction. [VERDICT] Planning must replace the myth of a stable versus unstable India with region-specific microzonation, engineering investigation, responsible resource assessment and ecosystem-sensitive land use. Why this earns marks: it interprets 'relative', compares four tectonic settings, integrates resource and risk consequences, and ends with a qualified policy verdict.

MUST-KNOW FACTS

1. Claim first.

2. Named region second.

3. Mechanism third.

4. Qualification and verdict last.

UPSC TRAPS

WRONG: Use the same evidence volume for every mark level.

CORRECT: Scale dimensions and examples to 10/15/20 marks.

MAINS ANGLE

it interprets 'relative', compares four tectonic settings, integrates resource and risk consequences, and ends with a qualified policy verdict.

STUDY LINK

Visual 16 answer spine.

End of Register Notes - UPSC Agent / Copilot CLI